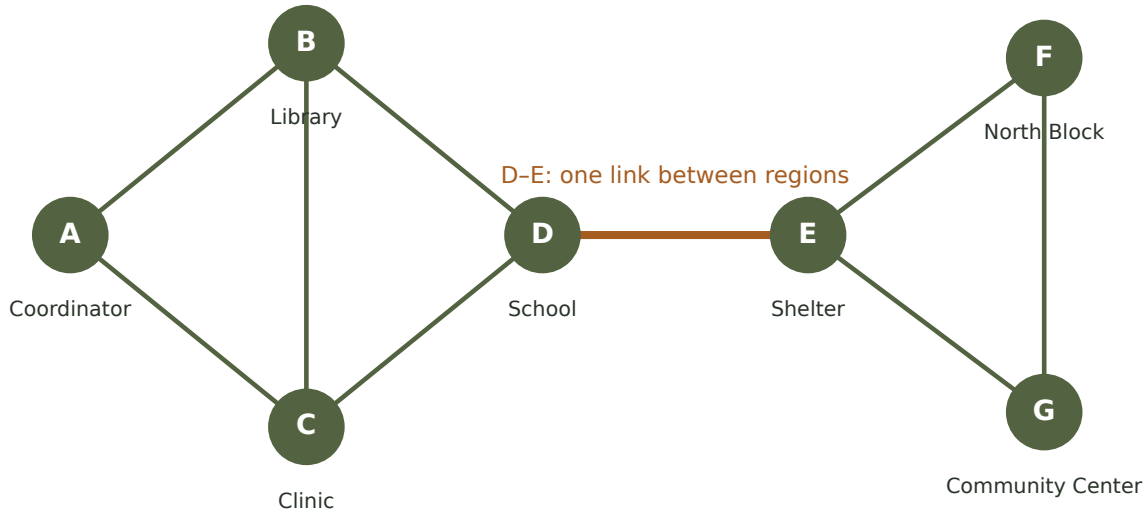
CHAPTER 9 / NETWORKS AND COLLECTIVE SYSTEMS

Networks, paths, and spreading

Explore food-sharing partners, walking routes, public alerts, and student groups to see how local connections shape whole systems.

Seven locations. Nine two-way communication links.



The drawing shows who can contact whom directly. It is not a geographic map.

The question for the week

A public alert may depend on a fragile communication link. A senior hub may depend on one food-sharing partner. A walk with fewer segments may take more time. What can network mathematics tell us about these choices?

The seven-location alert network provides continuity across the calculations. Different examples involving food-sharing, walking routes, social adoption, referrals, and neural signals let us compare what changes when a connection means something different. No earlier study of network theory is assumed.

Online reading · Network explorer · Guided laboratory

9.1 Different connections, different questions

Think about a food pantry coordinating with nearby organizations, someone walking to a hospital, and a county sending an emergency alert. Each situation depends on connections. A missing partnership, a slow route, or a failed communication channel can change what people are able to do.

A *network* records relationships. The dots represent people, places, or other objects, while the lines represent connections whose meaning we define. The same mathematics can describe direct contacts, routes, and what happens when a connection breaks.

We will develop the basic ideas using one county alert network with seven locations. That gives us a familiar reference for each new calculation. Along the way, we will also examine different networks for food access, travel time, student study groups, and referrals. These examples show when the mathematics transfers – and when the real setting changes its meaning.

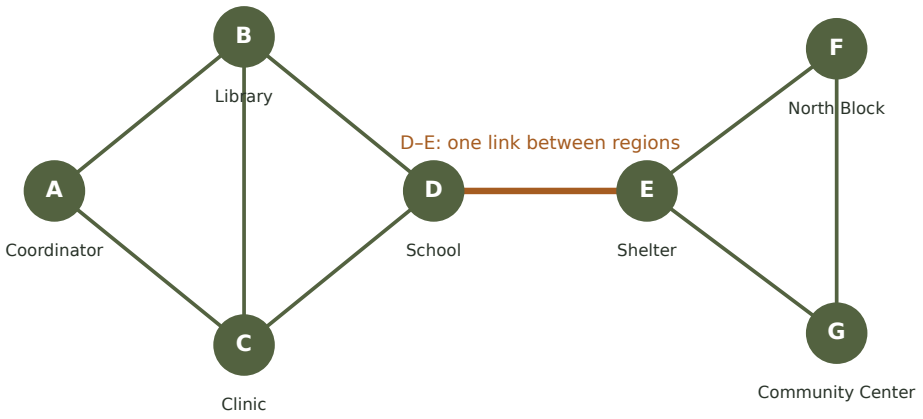
The network explorer uses the county network for controlled experiments: you can change a connection or the spreading rule and compare the consequences. Other examples in the chapter use different graphs.

9.2 Representing locations and links

Each dot in a network drawing is a *node*. A node represents one object we choose to study. Each line is an *edge*, or link, joining two nodes. In our alert model, a node is a location, and an edge means that the two locations have an established two-way communication channel.

Mathematicians also call a network a *graph* and a node a *vertex* (plural: vertices). This use of "graph" differs from a graph of temperature against time. A network drawing shows relationships; its horizontal and vertical positions need not be measured quantities.

Seven locations. Nine two-way communication links.



The drawing shows who can contact whom directly. It is not a geographic map.

Read the lines as available direct contacts. A can contact B and C directly. To contact G, A must use other locations as relays. The gold D–E link is the only connection between the two regions.

Node	Location	Direct neighbors
A	Emergency coordinator	B, C
B	Library	A, C, D
C	Clinic	A, B, D
D	School	B, C, E
E	Shelter	D, F, G
F	North Block	E, G
G	Community Center	E, F

How to read this mathematical world

- *Only drawn links count.* A and D may look close, but there is no direct A–D edge.
- *The drawing is not a map to scale.* A longer line does not mean a slower or longer trip. Moving a dot without changing its connections leaves this network unchanged.
- *A crossing is not automatically a junction.* You can transfer from one edge to another at a node. Lines that happen to cross without a marked node do not create another location.

We can record the same network with an *edge list*: AB, AC, BC, BD, CD, DE, EF, EG, FG. "AB" is shorthand for the connection joining A and B. Because the links work both ways, AB and BA describe the same edge, counted once.

Assumptions we will keep using

The main network is *undirected*: every link works both ways. It is *unweighted*: all links count equally. It is also *simple*: there are no links from a node to itself and at most one link per pair. Those assumptions determine which formulas apply.
The seven-location system is hypothetical. For a real emergency network, channel reliability, direction, and delays may also affect the result.

Pause and try 1

A community center can call a shelter, but the shelter cannot call back using that channel. Would an ordinary two-way line accurately represent this relationship?

9.3 Degree means the number of direct neighbors

The *degree* of a node counts the edges that touch it. In this simple network, that is also the number of its direct neighbors. "Degree" here is a count, not an angle or a temperature.

Example · Count the school's contacts

The edges touching D are BD, CD, and DE. Its neighbors are B, C, and E, so the degree of D is 3. We write $\deg(D) = 3$. The notation means "the degree of node D." A can eventually reach D, but A is not a direct neighbor and does not count toward D's degree.

Node	A	B	C	D	E	F	G
Degree	2	3	3	3	3	2	2

Count the same links from two viewpoints

Add the degrees: $2+3+3+3+3+2+2 = 18$. Yet the edge list has only 9 links. Why twice as many? Each link has two ends. AB contributes 1 to A's count and 1 to B's count.

sum of degrees = 2 × number of edges

This is the *handshake rule*. Imagine counting how many handshakes each person had: every handshake appears in two people's totals. With 9 edges, the degree sum must be $2(9) = 18$. This gives us a useful check on our counting.

We can also use the average from Week 3: the mean degree is $18/7 \approx 2.57$. No individual location needs to have degree 2.57. An average summarizes the group; every actual degree here is a whole number.

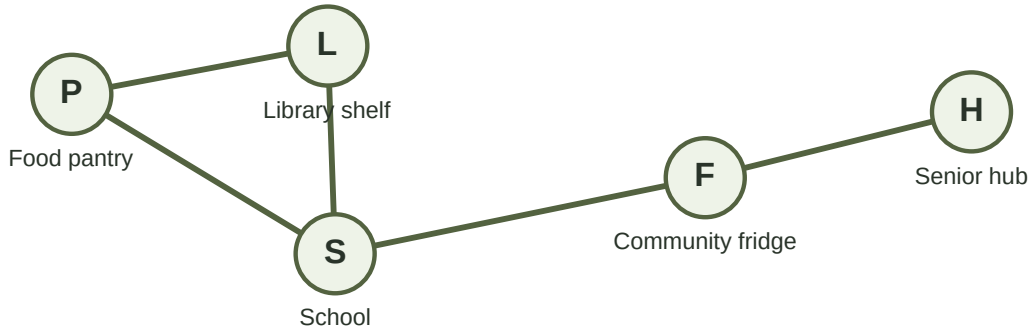
Another network · Neighborhood food-sharing partners

Suppose a food pantry (P), a library shelf (L), a school (S), a community fridge (F), and a senior hub (H) coordinate through five direct partnerships: PL, PS, LS, SF, and FH. Here an edge means two organizations can coordinate their work. It does not represent a road or guarantee that food is available.

The school connects directly to P, L, and F, so its degree is 3. The other degrees are 2, 2, 2, and 1. Together the degrees add to 10, twice the five partnerships. Among five organizations there are $5(4)/2 = 10$ possible pairs, so the density is $5/10 = 0.5$, or 50%. Notice that H has only one partner: the network's vulnerability depends on the arrangement as well as the number of connections.

A neighborhood food-sharing network

Edges PL, PS, LS, SF, FH — five different partnerships



Five links out of ten possible pairs: density = 5/10 = 50%.

This network has five organizations and five partnerships. The picture represents communication partnerships rather than physical distances or amounts of food.

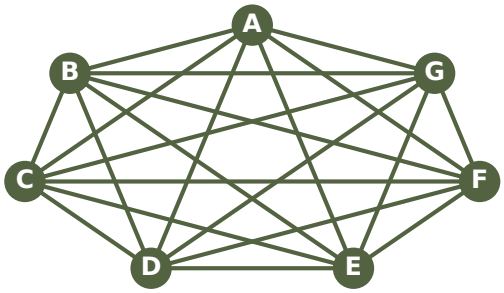
Pause and try 2

A different simple undirected network has 12 edges. What must the sum of all its degrees be? If you counted 23, what would that tell you?

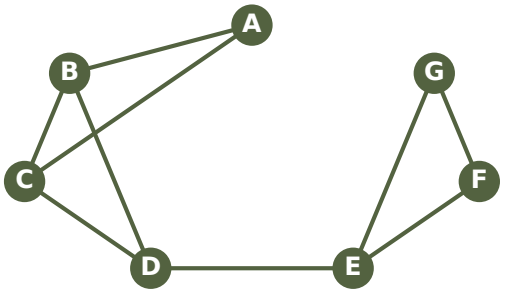
9.4 Density compares what exists with what could exist

Nine edges sounds like a lot until we ask how many are possible. Among seven locations, each one could have six other locations as neighbors. Counting $7 \times 6 = 42$ lists every pair twice: once as AB and again as BA. Divide by 2 to get 21 distinct possible links.

All possible pairs: 21 links



Links actually present: 9



Density = present ÷ possible = 9 ÷ 21 ≈ 0.429 = 42.9%

Keep the node count fixed when making this comparison. The network contains 9 of the 21 possible pairs. The extra line crossings in the complete drawing are not additional nodes.

The *density* of this simple undirected network is the fraction of possible links that are present:

$$\text{density} = \frac{\text{actual edges}}{\text{possible edges}} = \frac{9}{21} = \frac{3}{7} \approx 0.429.$$

As a percentage, that is about 42.9%. It means 42.9% of the possible *pairs of locations* have direct links. It does not mean that 42.9% of the locations are connected, or that any given message has a 42.9% chance of arriving. We have not introduced a probability of transmission.

Example · Read the general formula

Let n mean the number of nodes and m the number of edges. The letters are placeholders for counts. Then

$$\text{possible edges} = \frac{n(n-1)}{2}, \qquad D = \frac{m}{n(n-1)/2} = \frac{2m}{n(n-1)}.$$

For seven nodes and nine edges, substitute $n = 7$ and $m = 9$. You get $18/(7{\cdot}6) = 18/42 = 0.428571\ldots$. The letter D in the formula stands for density; it is separate from the location named D.

A density of 1 means every pair has a direct link. A density of 0 means no links exist. Adding a new link to our example raises density to $10/21 \approx 0.476$. The location of the new link matters: a new link inside one region and a new link between regions increase the count equally but can have different effects on reliability.

Pause and try 3

Five nodes have four undirected edges. How many distinct pairs are possible, and what is the density?

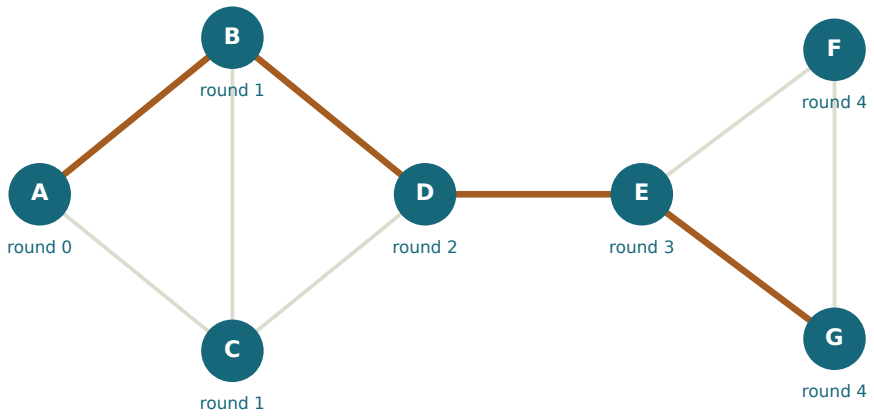
9.5 A path counts the links you actually use

A *path* is a route that follows edges from one node to another without repeating nodes. In an unweighted network, its *length* is the number of edges used. The *distance* between two nodes is the length of a shortest path.

Example · Get an alert from A to G

One route is $A \to B \to D \to E \to G$. Check each step against the edge list: AB, BD, DE, and EG all exist. The route names five nodes but uses four links, so its length is 4. The arrows here show the direction we travel along two-way links.
Another shortest route is $A \to C \to D \to E \to G$, also length 4. Several routes can share the same shortest length. A route that names A–D–G is invalid: neither AD nor DG exists in the original network.

Shortest-path distance from A = earliest arrival round



Highlighted route: A → B → D → E → G. Four links, so length 4.

Start at A with distance 0. B and C are one edge away; D is two away; E is three away; F and G are four away. These distance labels also become earliest arrival rounds under the spread rule introduced below.

Destination	A	B	C	D	E	F	G
Shortest distance from A	0	1	1	2	3	4	4

Connected does not mean directly connected

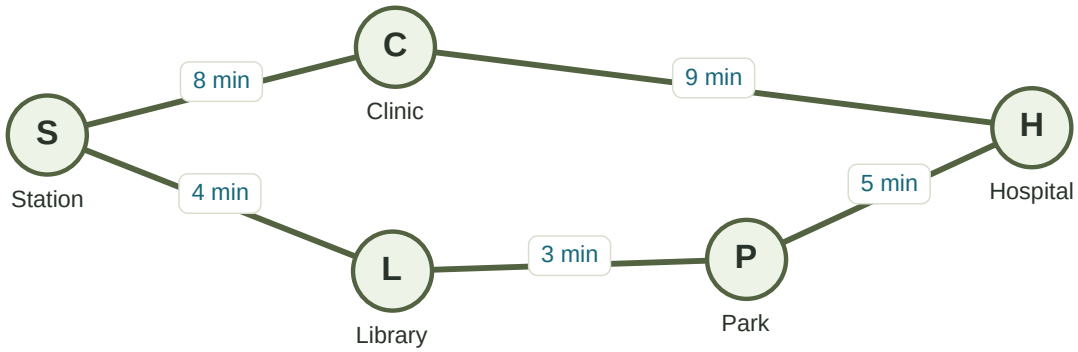
The network is *connected* because every pair of nodes can be joined by some path. A and G do not need a direct link for the whole network to be connected.

Another network · The quickest walk to a hospital

Imagine pedestrian paths between a station (S), clinic (C), hospital (H), library (L), and park (P). The direct walkways are SC, CH, SL, LP, and PH. The route S–C–H uses two links. The route S–L–P–H uses three. With each link counted equally, S–C–H is shorter.
Now add example walking times: SC takes 8 minutes and CH takes 9, totaling $8 + 9 = 17$ minutes. SL takes 4, LP takes 3, and PH takes 5, totaling $4 + 3 + 5 = 12$ minutes. The route with three links is faster. Before saying which path is shortest, we must decide whether we mean *fewest links* or *least travel time*.

A short walk is not always the quickest walk

Compare the fewest connections with the total travel minutes



Station–Clinic–Hospital: 2 links, 17 min. Via Library and Park: 3 links, 12 min.

These are hypothetical walking times. A real route choice would also consider accessibility, crossings, and individual walking speed.

A *connected component* is a whole group of nodes that can reach one another, with no path to nodes outside the group. Our original network has one component containing all seven nodes. If D–E is removed, there are two components: {A, B, C, D} and {E, F, G}. The braces simply mean "the group containing these nodes."

An isolated node, with no incident edges, is a component of one. If no path exists between two locations, their distance is *unreachable*, not zero. Zero is the distance from a node to itself.

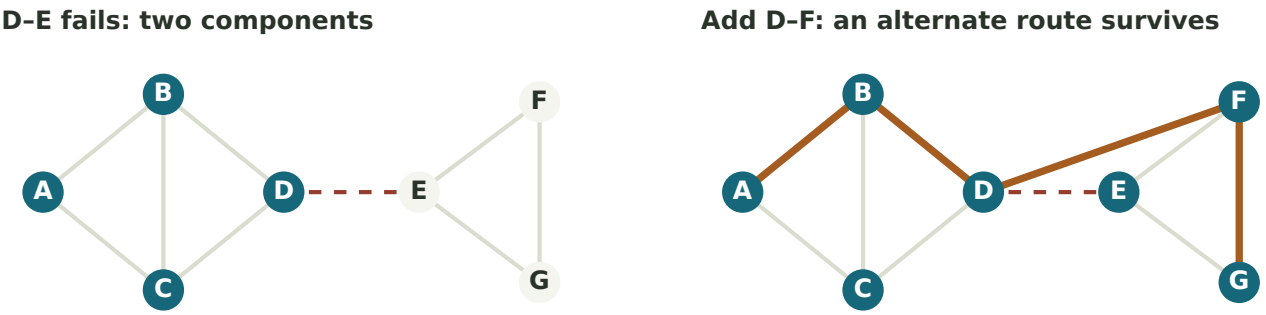
Pause and try 4

For edges PQ, QR, and ST, using nodes P, Q, R, S, T, how many components are there? What is the distance from P to R?

9.6 One vulnerable link can separate an entire region

A *bridge* is an edge whose removal increases the number of connected components. It need not look like a physical bridge. It is a link with a particular structural role.

D–E is a bridge: removing it changes our connected network from one component to two. By contrast, removing B–C does not disconnect the network. B can still reach C through A or D, so alternate routes remain.



Left: A cannot reach E, F, or G. Right: A → B → D → F → G still reaches G.

Dashed D–E is unavailable in both pictures. A new D–F link supplies the second route.

Test the failure explicitly. In the repaired drawing, A can reach G through A–B–D–F–G without using D–E. The dashed line represents the failed link, not a usable connection.

Example · Spend the budget on an alternate route

Suppose the county can add one new link. Adding A–D shortens a local route but leaves D–E as the only cross-region link. Adding D–F creates a second way into the right region. If D–E fails, the route A–B–D–F–G still works.

This is *redundancy*: another route can serve when one link is lost. The new network is more resilient to that failure. In fact, with D–F added, no individual edge is a bridge; every edge has an alternate route around it.

A link failure and a location failure are different tests

Removing a node also removes every edge touching it. A node whose removal increases the number of components is an *articulation point*. D and E are articulation points in the original network.

Adding D–F protects against the loss of D–E, but both cross-region routes still depend on D. If the school itself goes offline, B and C lose both connections through D. The repaired network is still vulnerable to the loss of that node. A claim about resilience should name the failure being tested.

Pause and try 5

In the three-node path P–Q–R, which edges are bridges? Which node is an articulation point?

9.7 "Important" can mean several different things

B, C, D, and E all have degree 3. Yet they do different jobs. B and C have alternate routes around them. D and E sit at the only passage between regions. Their equal degrees do not imply equal vulnerability or equal roles.

Idea	What you measure	Question it helps answer
Degree	Number of direct neighbors	Who can contact many others in one step?
Closeness	Shortest-path distances to the other nodes	Who is relatively few steps from everyone?
Betweenness	How often shortest paths between other nodes pass through a node	Who lies along many short routes between groups?
Bridge or articulation role	Whether removing a link or node splits a component	Where can one failure separate the system?

You do not need a betweenness formula this week. Trace routes between the two regions: every such path must pass through D and E. That gives a reason for their structural importance beyond counting neighbors. A bottleneck position is not a measurement of a person's worth or the value of a neighborhood.

Extension · Turn "few steps away" into an average

From A, the distances to the six *other* nodes are 1, 1, 2, 3, 4, 4. Their sum is 15. The average distance is $15/6 = 2.5$ edges. From D, the distances to A, B, C, E, F, G are 2, 1, 1, 1, 2, 2, whose sum is 9 and average is $9/6 = 1.5$. So D is closer on average to the rest of this network.
One common *closeness centrality* score reverses that average: $(n-1)/(\text{sum of distances})$. Thus A scores $6/15 = 0.400$, while D scores $6/9 \approx 0.667$. Larger scores mean shorter average distances. The explorer displays a dash if a node cannot reach the whole network; this particular formula assumes all those distances exist.

Another optional score, *degree centrality*, divides degree by the largest possible degree, $n-1$. For D this is $3/6 = 0.5$: D is directly linked to half of the six other locations. It describes a fraction of possible neighbors, not a probability of success.

Pause and try 6

If the objective is to reach the school directly from A, should you add A–D or D–F? If the objective is surviving loss of D–E, does the answer change?

9.8 An adjacency table answers "Is there a direct link?"

Adjacent means directly connected by an edge. An *adjacency matrix* is simply a square table with one row and one column for each node. A 1 records a direct connection. A 0 records its absence.

Choose the row for the starting node and the column for the other node. At row B, column D, you find 1 because BD is an edge. At row A, column D, you find 0, even though A can reach D using two links. The table records direct adjacency, not every possible path.

From / to	A	B	C	D	E	F	G
A	0	1	1	0	0	0	0
B	1	0	1	1	0	0	0
C	1	1	0	1	0	0	0
D	0	1	1	0	1	0	0
E	0	0	0	1	0	1	1
F	0	0	0	0	1	0	1
G	0	0	0	0	1	1	0

Look at three features. The diagonal from top left to bottom right is all zeros because no node links to itself. The entries mirror one another across that diagonal because the links work both ways. The sum of a row is its node's degree: D's row has three 1s.

Why learn a second representation?

A drawing helps us see clusters and routes. A table lets us check entries systematically and store a network in a computer. They describe the same connections. You can read the matrix as an ordinary table; matrix algebra is not required.

Pause and try 7

Does the 0 at row A, column G mean an alert can never get from A to G?

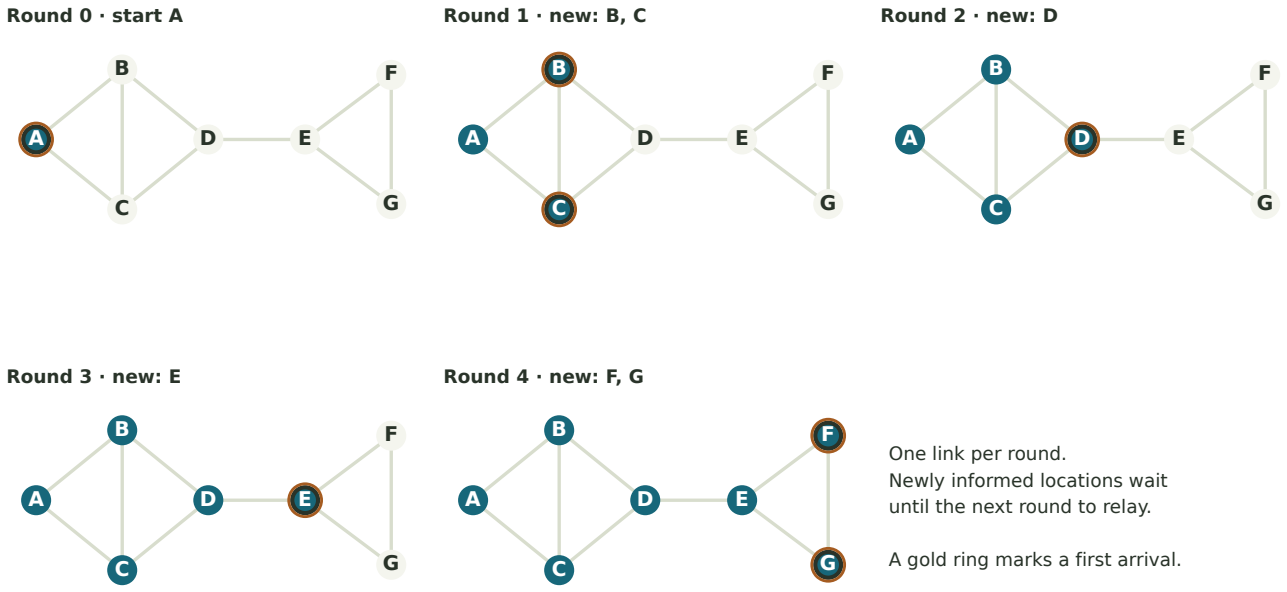
9.9 A network shows the routes. A rule determines the spread.

Diffusion means something spreading through connections. To calculate it, we need more than a picture. We need an initial state and a rule for updating that state, just as we needed a starting balance and an interest rule in Week 6.

The alert rule, stated precisely

1. At round 0, only A has the alert. Round 0 is the starting state, before any sending.
2. During each round, every location that was informed *at the start of that round* sends to all its direct neighbors.
3. New recipients may begin sending in the following round. All updates use the same previous snapshot.
4. Once informed, a location stays informed. Every available transmission succeeds.

One round is one model time step. We have not said it equals a minute. That would require information about real transmission times.



Read "new" as first arrivals in that round. B and C receive the alert together in round 1. They wait until round 2 to relay it onward. Nothing travels across two links within one round.

Round	Newly informed	All informed	Total
0	A (the initial source)	A	1
1	B, C	A, B, C	3
2	D	A, B, C, D	4
3	E	A, B, C, D, E	5
4	F, G	A, B, C, D, E, F, G	7

Round 1 has 2 newly informed nodes and 3 informed nodes in total. Those are different counts. If B and C both send to D, D is still only one new recipient. We count locations, not message copies.

Under this exact rule, a node first receives the alert at the round equal to its shortest-path distance from A. A four-link path needs at least four rounds, and sending along a shortest path achieves that time. With D–E removed, no number of rounds can deliver A's alert to F or G because no path remains.

Real alerts can be delayed, ignored, misunderstood, or blocked by an outage. The model's timing is conditional on the rule. A deterministic spread calculation does not establish a real-world arrival guarantee.

Pause and try 8

At round 2, how many nodes are newly informed and how many are informed in total? If B and C both contact D, why do we not count two new nodes?

9.10 Sometimes one informed neighbor is not enough

An emergency relay might pass on an alert immediately. A person deciding whether to adopt a new practice may wait for support from several neighbors. A *threshold model* gives a simple way to describe that different situation. "Active" now means a location or person has adopted the behavior; it does not merely mean a message arrived.

Use this rule: an inactive node becomes active in the next round if *at least half* its neighbors are active now. Active nodes stay active. Starting active nodes are called *seeds*. We again update all nodes from the same previous snapshot.

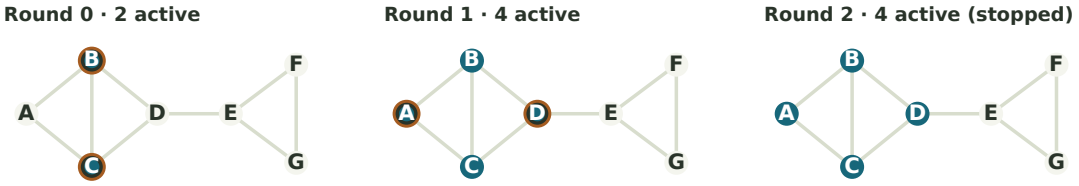
Number of neighbors	1	2	3	4	5	6
Minimum active neighbors needed	1	1	2	2	3	3

For 3 neighbors, half is 1.5. Since we cannot have 1.5 active neighbors, at least 2 must be active. The fraction 2/3 passes one-half, while 1/3 does not. For an isolated node, this model leaves it inactive unless it is a seed; there is no neighbor influence to observe.

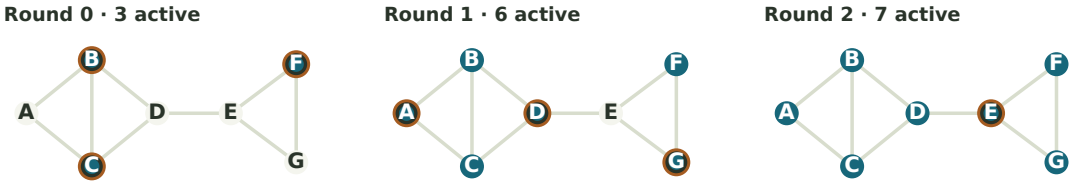
Example · Why the cascade stops

Start with B and C active. In the next round, A activates because both its neighbors are active; D activates because 2 of its 3 neighbors are active. The active set is now A, B, C, D.
E has neighbors D, F, G. Only D is active, giving $1/3 < 1/2$, so E waits. F and G have no active neighbors. No one new activates in the following round. The process stops with four active nodes, although the network is connected.

Seeds B, C: the cascade stops before E



Add F as a seed: the cascade reaches all seven locations



Top: the half-neighbor threshold blocks spread at E. Bottom: F is also a seed, so G activates in round 1. At the start of round 2, E sees active D, F, and G, so E activates. Gold rings mark seeds or new activations.

Example · One extra seed changes the outcome

Start instead with B, C, and F active. In round 1, A and D activate as before. G also activates: its neighbors are E and F, so the one active neighbor F reaches the half-neighbor threshold. E still waits in round 1 because only F was active among its neighbors at round 0.
At the start of round 2, E has three active neighbors: D, F, G. E activates, so all seven nodes are now active. The network and the rule stayed the same; adding F to the starting set changed the collective outcome.

A *cascade* is a sequence in which one activation helps trigger later activations. This is one meaning of a *collective effect*: the result belongs to interactions across the group. A cascade can stop, spread partly, or reach everyone. It is not automatically good or bad; the meaning depends on what is spreading.

The half-neighbor rule is an invented model, not a measured law of human behavior. People may use different thresholds, and adoption can reverse. Its value here is that we can state it precisely and see what follows.

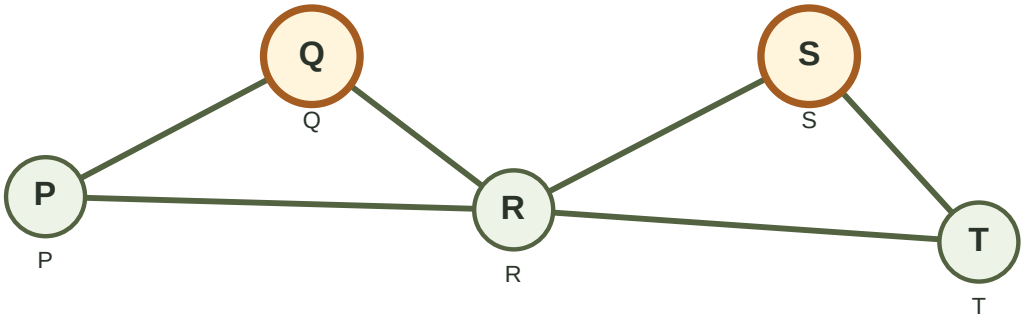
A different social network · Joining a study group

Five students, P, Q, R, S, and T, talk directly along PQ, PS, QR, RS, RT, and ST. Imagine that an undecided student joins a study group if at least half of their contacts have joined. With only Q participating at the start, P joins in round 1 because Q is one of P's two contacts. R sees only one of three contacts participating, and T and S see none. The process stops with P and Q.
If Q *and* S start together, P joins in round 1 because Q and S are both of P's contacts, R joins because Q and S are two of three, and T joins because S is one of two. All five participate after one

round. This is a second network with a different arrangement of connections, and it shows how the initial group matters. The rule is a mathematical experiment, not a prediction of anyone's actual decision.

Study-group participation: two starting conditions

Edges PQ, PR, QR, RS, RT, ST; gold rings mark initial participants



With half-neighbor thresholds, Q alone stalls; starting with Q and S activates all five.

Compare the two starting conditions: Q alone, then Q and S. With the half-neighbor rule, the second starting group reaches everyone in one update.

Pause and try 9

A node has five neighbors and a 60% threshold. How many active neighbors does it need? Does that tell you whether the entire network will activate?

9.11 Arrows and weights add information

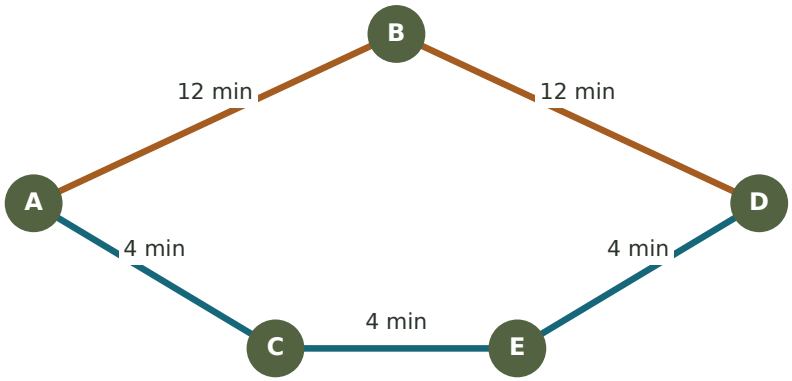
A *directed* edge is drawn with an arrow. If $A \to B$ means "A can send to B," it does not also mean that B can send to A. For $A \to B \to C$, A can reach C but C cannot return to A using those arrows. A path must obey every direction.

Out-degree counts outgoing arrows; *in-degree* counts incoming arrows. In that chain, B has one incoming and one outgoing arrow. These describe different roles. We must define what an arrow means before interpreting its direction.

For another example, a clinic (C) might send a referral to a specialist (S) or a laboratory (L); either one can forward a report to a care coordinator (R). The directed connections are $C \to S$, $C \to L$, $S \to R$, and $L \to R$. The clinic has out-degree 2, and the coordinator has in-degree 2. A report can follow $C \to S \to R$, but those arrows do not establish a route back to C. This is a simplified workflow, not a model of every clinical responsibility.

A *weight* is a number attached to an edge. It might represent minutes, dollars, capacity, or a multiplier. Its meaning determines how we use it.

Fewest links and least time answer different questions



A–B–D: 2 links, 24 minutes. A–C–E–D: 3 links, 12 minutes.

For this travel-time model, add the minutes along a route. A–B–D uses fewer links, but A–C–E–D takes less time. The line lengths on the drawing do not measure the travel times.

In the AI application, a weight means a multiplier applied to a numerical signal. A negative weight can subtract from the next node's sum. That is different from a negative travel time, which would not fit the road example. A weight has meaning only in its mathematical setting.

9.12 Structure, updates, and choices belong together

The network is the *structure*: who can interact with whom. The current informed or active set is the *state*. The spread or threshold rule creates *dynamics*: how that state changes over successive rounds. These words describe things you have already calculated.

Week 5's feedback systems combined observations with responses. A network adds the routes through which those observations and responses can travel. More sharing can produce more recipients, who then share further: an amplifying feedback process. A failed communication link can prevent a corrective message from arriving at all.

Example · A constrained design decision

Weeks 7–8 asked us to choose an objective and describe allowed choices. Here the constraint is "add one new link." If the objective is direct contact from A to D, A–D succeeds. If the objective is continued access across regions after D–E fails, D–F succeeds. Each adds one link and gives density 10/21, but the consequences differ.

We could test all allowed links for a chosen criterion. That would optimize the stated model; it would not decide what a community ought to value. Costs, staffing, accessibility, trust, privacy, and unequal outage risks may change the real decision.

A practical routine for a new network

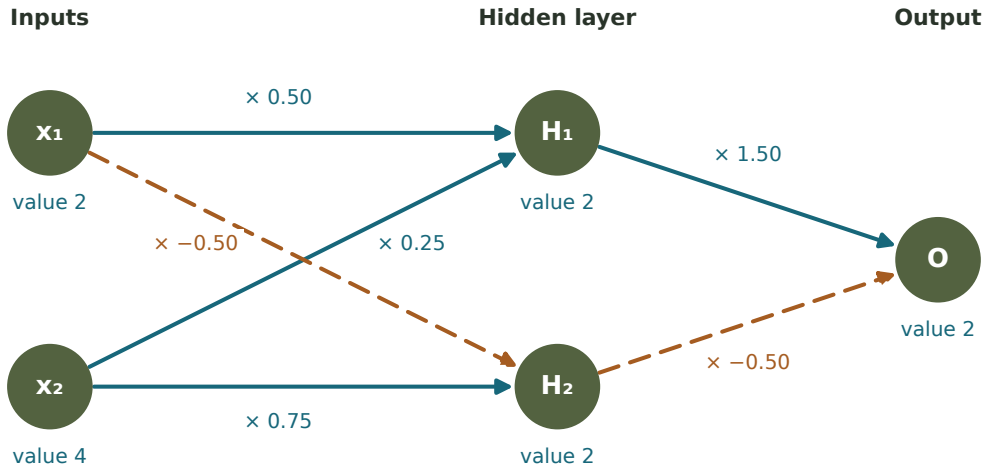
1. State what each node and edge represents, including direction and weights.
2. Write an edge list or a neighbor table so the drawing can be checked.
3. Choose a question: direct contacts, travel, vulnerability, or spread.
4. Calculate with the appropriate rule and show a route, count, or sequence of states.
5. Test a proposed change against the original objective.
6. Explain one feature the model leaves out and what further information would help.

For Week 10, keep this habit: a precise local rule can generate a result that is hard to guess from a quick glance. Networks gave us cascades across locations. Next week, repeated number and shape rules will give us other surprising patterns.

9.13 AI connection: signals along weighted arrows

Imagine a community center testing a small program that combines two indoor sensor scores. The heat score is $x_1 = 2$ and the motion score is $x_2 = 4$. The subscripts identify two different inputs. These invented scores are rescaled to numbers from 0 to 6; they are not temperatures or probabilities.

The diagram is a tiny *linear signal network*. Every arrow has a multiplier called a weight. Each receiving node adds its incoming weighted signals. The intermediate nodes H_1, H_2 form a hidden layer: a set of calculations between the inputs and output. "Hidden" means intermediate, not secret.



At each receiving node: multiply along each incoming arrow, then add.

Worked calculation: multiply, then add

At the first hidden node,

$$H_1 = 0.50x_1 + 0.25x_2 = 0.50(2) + 0.25(4) = 1 + 1 = 2.$$

At the other hidden node, one contribution is negative:

$$H_2 = -0.50x_1 + 0.75x_2 = -0.50(2) + 0.75(4) = -1 + 3 = 2.$$

Now send those values through the two arrows to the output:

$$O = 1.50H_1 - 0.50H_2 = 1.50(2) - 0.50(2) = 3 - 1 = 2.$$

The sign of a weight changes a contribution; the arrow determines the direction. A weight is a multiplier, not a transmission probability.

A connected input can have no net effect

Hold x_1 fixed and increase x_2 by 1. The path through H_1 adds $0.25(1.50) = 0.375$ to the output. The path through H_2 adds $0.75(-0.50) = -0.375$. The effects cancel exactly. Changing the motion score changes both hidden values but leaves the output unchanged.

In fact, substituting the hidden formulas into the output gives

$$O = 1.50(0.50x_1 + 0.25x_2) - 0.50(-0.50x_1 + 0.75x_2) = x_1.$$

This algebra is an extension; the two path contributions already explain the cancellation. Connections show where influence *can* travel. The weights determine how those influences combine.

Remove one edge and test again

With the original inputs 2 and 4, remove the arrow $x_2 \to H_1$. The new hidden values are $H_1 = 0.50(2) = 1$ and $H_2 = 2$. Thus $O = 1.50(1) - 0.50(2) = 0.5$. The other motion-score path remains, so the earlier cancellation is broken.

An activation rule adds another transformation

The original model only multiplies by constants and adds. Combining such linear calculations still gives a linear relationship. Neural networks commonly add *activation functions*, rules that transform a node's sum before sending it onward.

For example, ReLU keeps a nonnegative sum and replaces a negative sum by zero. With $x_1 = 4, x_2 = 0$, the raw hidden sums are 2 and -2. Without the extra rule, the output is $1.50(2) - 0.50(-2) = 4$. With ReLU at the hidden nodes only, the sent values are 2 and 0, so the output is 3. This introduces a nonlinear relationship. Real models also commonly add adjustable constants called biases; our model sets them to zero.

A forward pass is different from training

A *forward pass* uses current inputs and weights to calculate an output. Training uses examples with targets, compares outputs with those targets, and changes parameters in response to errors. That makes training a feedback process.

For the original prediction 2 and an invented target 3, define signed error as target minus prediction: $3-2 = 1$. If we change only the H_1-to-output weight from 1.50 to 2.00, the new output is $2.00(2)-0.50(2) = 3$. This hand-chosen change matches one target; it does not establish accuracy on other examples. Backpropagation and an optimization method organize parameter updates in larger models.

The weights, scores, and targets here are invented and have not been fitted to sensor data. The output is a toy score. The online AI application lets you change inputs, remove an edge, and compare the two node rules.

Chapter Review

For questions about the county alert network, use nodes A–G and edges AB, AC, BC, BD, CD, DE, EF, EG, FG. Other questions provide different networks. Unless a problem says otherwise, links are two-way. Show a count, route, or update sequence and interpret the result in its setting.

1. Find all seven degrees. Check the sum against the number of edges, then calculate the mean degree.
2. Five food-sharing partners P, L, S, F, H have direct two-way partnerships PL, PS, LS, SF, FH. Find the degrees of S and H, the number of possible pairings, and the density. What does that density say about the partnerships?
3. Find two different shortest paths from A to G. State their lengths. Find the distances from B to F and from D to every other node.
4. What are the connected components after DE is removed? Is the distance from A to F then zero, very large, or unreachable? Explain.
5. In the food-sharing network PL, PS, LS, SF, FH, which edges are bridges and which nodes are articulation points? Describe what losing the community fridge F would do to the senior hub H.
6. Add DF. Give an A-to-G route that survives loss of DE. Does this addition also protect against loss of D itself?
7. Compare adding AD with adding DF. With all links working, at which round is everyone informed from A under automatic sending? Which proposal meets the goal of surviving failure of DE?
8. In the adjacency table, find row A/column D, row D/column E, and row E/column E. What is the sum of row D? Explain why a zero at A/G does not mean G is unreachable from A.
9. A alone is informed at round 0. List the newly informed and total informed nodes after rounds 1–4 under the automatic rule. Explain why two senders contacting D do not count as two new locations.
10. Five students P, Q, R, S, T have conversation links PQ, PS, QR, RS, RT, ST. Each joins a study group if at least half of their neighbors have joined. Starting with Q alone, list the active students after each round until the process stops. Now start with Q and S. What happens after one round?
11. A node has five neighbors and a 60% threshold. How many active neighbors are required? Does that local count determine the outcome of the whole network?
12. A clinic C refers to specialist S and laboratory L; both may send a report to coordinator R. The directed links are $C \to S$, $C \to L$, $S \to R$, $L \to R$. Find C's out-degree, R's in-degree, and whether R can return to C along the same arrows. In the walking network, compare a two-link walk of 8+9 minutes with a three-link walk of 4+3+5 minutes.
13. For the AI network with inputs 2 and 4, find H_1, H_2, O. Remove $x_2 \to H_1$ and recalculate. Why does changing x_2 have no net effect in the original sum-only model?
14. With target 3 and prediction 2, use signed error = target minus prediction. Change only the H_1-to-output weight to 2.00 and recalculate. What does matching this one target establish, and what does it leave unknown?
15. Choose a real network. Define nodes, edges, a useful question, and one limitation of a simple undirected, unweighted representation.

A reliable explanation has four parts

State the model and question. Show a route or calculation. Interpret the result in the setting. Name an assumption that limits the conclusion.

Solutions and Further Reading

Short check 1. No. Use a directed arrow from the community center to the shelter, or explain a different model. A two-way line would incorrectly permit travel in both directions.

Short check 2. The sum must be 24. A total of 23 signals a counting or recording error under these assumptions. Every edge contributes exactly 2.

Short check 3. There are $5(4)/2 = 10$ possible pairs. Density is $4/10 = 0.4$, or 40%. Count possible pairs before taking the fraction.

Short check 4. There are two components: {P,Q,R} and {S,T}. The path P–Q–R has length 2. There is no path from P to S.

Short check 5. Both PQ and QR are bridges: removing either isolates one end. Q is the articulation point: removing it leaves P and R separate. A bridge does not have to be the "middle" edge of a longer path.

Short check 6. Add A–D for direct A-to-school contact. Add D–F for an alternate cross-region route. "Best" depends on the objective, just as in Weeks 7–8.

Short check 7. No. It says there is no direct A–G link. The four-edge route A–B–D–E–G still exists. Adjacency and reachability answer different questions.

Short check 8. D is the one new node; four nodes are informed in total. Two messages to the same location do not create two distinct informed locations.

Short check 9. It needs 3 because 60% of 5 is 3. That is only a local requirement. The network structure, the other thresholds, and the starting seeds determine whether a larger cascade occurs.

Chapter Review: solutions and reasoning

1. In order A–G, degrees are $2, 3, 3, 3, 3, 2, 2$. Their sum is 18, which equals $2(9)$. The mean is $18/7 \approx 2.57$ neighbors.

2. S connects to P, L, F and has degree 3, while H connects only to F and has degree 1. Among five nodes, $5(4)/2 = 10$ pairs are possible. Five partnerships give density $5/10 = 0.5$, or 50%. Density counts direct partnerships, not the amount of food supplied.

3. A–B–D–E–G and A–C–D–E–G each use 4 edges. B-to-F distance is 3 via B–D–E–F. From D to A,B,C,E,F,G the distances are $2, 1, 1, 1, 2, 2$; their sum is 9 and mean is 1.5.

4. The components are $\{A, B, C, D\}$ and $\{E, F, G\}$. F is unreachable from A. Waiting cannot create a missing path. Zero is the distance from A to itself.

5. SF and FH are bridges. S and F are articulation points. If F stops coordinating, H loses its only partner, though P, L, and S remain connected. An edge test removes one relationship; a node test removes an organization and every link touching it.

6. A–B–D–F–G survives loss of DE. But removing D itself leaves $\{A, B, C\}$ separate from $\{E, F, G\}$ even with DF added. Both cross-region links depended on D.

7. AD gives complete delivery by round 3 with all links working; DF gives complete delivery by round 4. DF, unlike AD, supplies a cross-region route after DE fails. Both give 10 edges and density $10/21$ before failures.

8. The entries are 0,1,0; row D sums to 3. A zero at A/G only rules out a direct link; paths using intermediate locations still exist.

9. New nodes by rounds 1–4: $\{B, C\}, \{D\}, \{E\}, \{F, G\}$. Cumulative totals: 3,4,5,7, including the initial A. Multiple message copies arriving at D inform the same one location.

10. With Q alone, P joins in round 1 because Q is one of P's two neighbors. R sees one active neighbor out of three, and S and T see none, so the active set stops at $\{P, Q\}$. With Q and S initially active, P sees both of two, R sees two of three, and T sees one of two. All five students are active after round 1.

11. Three active neighbors, since $0.60(5) = 3$. The outcome also depends on other thresholds, the network, and seeds.

12. C has two outgoing referrals, R has two incoming reports, and R cannot return to C without a reverse arrow. The two-link walk takes $8 + 9 = 17$ minutes; the three-link walk takes $4 + 3 + 5 = 12$ minutes. Fewest connections and least travel time need not agree.

13. $H_1 = 2, H_2 = 2, O = 2$. After removal, $H_1 = 1, H_2 = 2, O = 0.5$. The original two paths from x_2 contribute $+0.375$ and -0.375 per input unit, so their net effect is zero.

14. Error is 1. The new output is $2.00(2) - 0.50(2) = 3$, giving error 0 for this example. It matches one target; performance on other examples remains untested.

15. Answers vary. For a transit network, nodes might be stops and edges direct routes between stops. A useful question is whether a closure separates a neighborhood. An unweighted graph omits waiting and travel times; an undirected graph omits one-way service.

Optional background

- NIST, *Graph* and *Bridge*, Dictionary of Algorithms and Data Structures.
- Google Machine Learning Crash Course, *Nodes and hidden layers* and *Activation functions*.

All networks, sensor weights, and scenarios in the exercises are constructed teaching examples. The mathematical calculations follow their stated assumptions.